

A Macroeconomic Approach to Optimal Fertility Across Nations

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Abstract

Should countries aim for the replacement level of fertility? In other words, is the optimal fertility rate below or above the replacement level? Does the answer differ across countries, and if so, why? This paper explores these questions through the lens of a dynastic model that captures two opposing channels through which fertility affects the long-run national income and welfare. On the one hand, higher fertility generates positive externalities, as a larger population contributes to a greater stock of ideas and a larger labor force in the long run. On the other hand, negative (fiscal) externalities arise from the child penalty and the quantity–quality trade-off, which reduce aggregate labor input and increase the equilibrium tax rate. Using data from nine high-income countries, I calibrate and estimate key parameters governing preferences, technology, and government spending to match cross-country differences in fertility, education, leisure, public expenditure, and income levels. Solving a planner’s problem for each country, I find that the optimal fertility rate remains below replacement rate in many countries. This is particularly true for countries such as Germany, Japan, and Spain, where higher child penalties raise the social costs of increased fertility through the associated output losses.

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